OCC: SOFTWARE ENGINEER YEAR 2

SAQA ID

NQF 6

Software design and testing C#

SDT621

SUMMATIVE ASSESSMENT

2026



CTU training solutions | Private Tertiary Education

Table of Contents

DECLARATION OF AUTHENTICITY....3
QUALIFICATION AND ASSESSMENT DETAILS....4
Instructions:....5
Submission Requirements....5
Evidence Collection Requirements....5
To support your assessment submission, you must:....5
GitHub Submission Requirement....5
Summative Assessment Criteria:....6
Example of Collecting Evidence for your submission pdf:....6
SECTION A — Knowledge Questions (40 Marks)....7
True / False Questions [10]....8
SECTION B — Short Practical Questions (10 Marks)....9
Question 1....9
SECTION C: Scenario-Based Practical [50 Marks]....10
Question 1....10
SECTION C.2 — Button Event Logic Implementation (20 Marks)....10
SECTION C.3 — Software Testing Report (15 Marks)....11
SECTION C.4 — GitHub Version Control Evidence (5 Marks)....11
SECTION A — Knowledge Questions (40 Marks)....12
SECTION B — Short Practical Questions (10 Marks)....13
SECTION C — Scenario-Based Practical (50 Marks)....13
SECTION C.2 — Button Event Logic Implementation (20 Marks)....14
SECTION C.3 — Software Testing Report (15 Marks)....14
SECTION C.4 — GitHub Version Control Evidence (5 Marks)....14
SECTION C Mapping Summary....14

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DECLARATION OF AUTHENTICITY

I Wian Sandenbergh

hereby declare that the contents of this assessment
are entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work not accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

I have read and understood the assessment requirements and the Examination Rules and Regulations.

Signature: WS

Date: 5/27/2026

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QUALIFICATION AND ASSESSMENT DETAILS

Programme Title	OCC: SOFTWARE ENGINEER
Module Title	Software design and testing C#
Module Code	SDT621
NQF Level	6
Assessment Type	SUMMATIVE ASSESSMENT
Hand Out Date	27 / 0 5 / 2026
Hand in Date	27 / 05 / 2026
Total Marks	100
Assessment Lead Developer	Lusukama Selemani
Internal Moderator	Faith Muwishi

Student Name and Surname	**Wian Sandenbergh**
Student Number	**20240114**

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Instructions:

Test or Debug Source Code to Ensure Client Needs Are Met

Please read the following instructions carefully before attempting this assessment:

- Read each question carefully and consider the mark allocation before answering.
- Answer all questions in the assessment.
- Provide clear, well-structured C# code and explanations where required.
- Apply your understanding of software testing and debugging techniques when solving the given scenarios.
- Ensure that your solutions meet the client requirements described in the scenario.
- Use meaningful variable names, proper formatting, and correct programming logic.
- Test your application before submission to confirm that it runs correctly.

Submission Requirements

- For your final submission, you must include:
- A signed Declaration of Authenticity (if not already included in the assessment document provided)
- Your completed Answer Sheet (PDF document) (preferably the official template provided with this assessment)
- Your Visual Studio project files (compressed as a ZIP file)

Evidence Collection Requirements

To support your assessment submission, you must:

- Take a screenshot of the program output for each question
- Ensure each screenshot shows the full screen, including the system date and time
- Ensure screenshots clearly demonstrate correct program execution
- Do NOT submit screenshots of your source code
- Submit your complete Visual Studio project folder as a ZIP file

GitHub Submission Requirement

You are required to:

- Push your project code to GitHub
- Ensure the repository contains the correct and complete solution
- Share the GitHub repository link in your submission for verification and review
- Example: https://github.com/your-username/project-name

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Summative Assessment Criteria:

Applied Knowledge

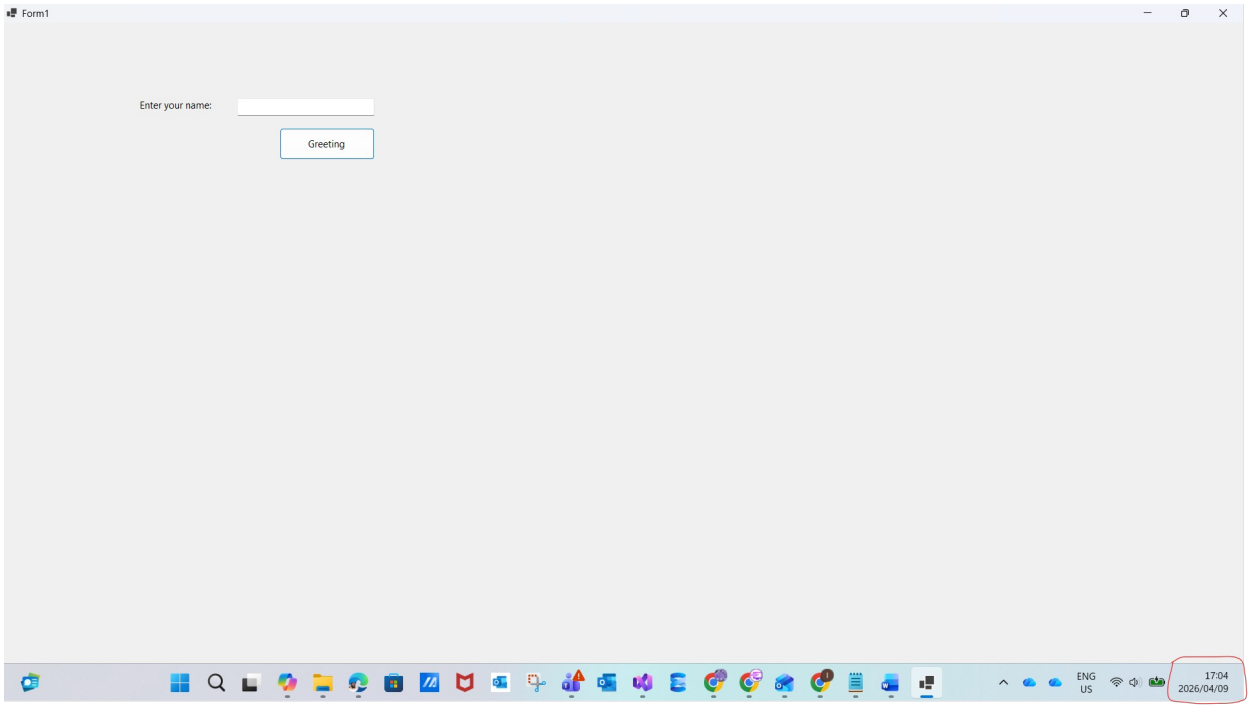
- AK0101 Knowledge of application testing techniques
- AK0102 Knowledge of test report writing

Internal Assessment Criteria

- IAC0101 functional and non-functional tests are identified
- IAC0102 The correct tests are applied
- IAC0103 A test report is produced

Example of Collecting Evidence for your submission pdf:

Question1 :



Ensure each screenshot shows the **full screen**, including the **system date and time**

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SECTION A: Knowledge Questions (40 Marks)

Question 1 (30 Marks)

1.1 Define non-functional testing and explain what type of system qualities it evaluates. (2)

Non-functional testing is testing features of the program that doesn't affect the core definement of the program.

1.2 Explain load testing as a non-functional testing type. (1)

Load testing makes sure the program can still run smoothly when put under a lot of stress. This is not needed for an program but it does help make it easier to use.

1.3 Describe the difference between load testing and stress testing in non-functional test design. (2)

Load testing is making sure that the program is optimised to run functions fast, stress testing is making the program run as many data as possible to see how much it can handle before it breaks

1.4 Explain the difference between black-box testing and white-box testing. (3)

Black box testing focuses on the testing of system behaviour without focusing on the inner code structure. White-box testing focuses on the inner code structure and the integrity of the logic.

1.5 Describe equivalence partitioning and explain how it improves functional testing efficiency (3)

This ensures that all tests get sorted into

1.6 A Loan Eligibility Processing System accepts customer income values between:

R1 000 and R150 000 Design TWO boundary value test cases to verify correct system behaviour. (4)

I'd create one outside of the values specified that it can handle and then also on in the middle of it. This will help me test the output of a system when I try to use it and the other test case is when I try to break it.

1.7 Define error-case testing. (1)

Error-case testing is making sure that your program doesn't give errors when your give it wrong information or something that can break/cause a problem.

1.8 Explain why testing unexpected user inputs is important when validating software reliability. (2)

The user might be new to the program or have no understanding on what to enter, if your input in case sensitive and the user enters everything in lower the won't know the difference because they type out the word. You also don't want wrong data in your program this might cause errors or and cause problems in your database.

1.9 Provide TWO examples of error-case tests suitable for a login system. (2)

If I enter the wrong password will it promp the user to try again that something is wrong. If I enter no values and try to login will it prompt me to enter values or throw an error because the code never got any inputs.

1.10 Define a test case in software testing. (2)

a Test case is a test where you see if your program can run and complete the test to make sure its core functionality works as intended.

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1.11 List THREE components typically included in a structured test case. (3)

TestCase, Expected result, Status

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1.12A testing session executed:

Test Cases	Passed	Failed
100	80	20

Prepare a structured test summary section for a system testing report including:

Test results overview

Recommendation to stakeholders (5)

As we can see our program has passed 80% of all tests we did making the total amount of work left on the program to fix less. We only need to fix 20% of the test cases used on the program putting our program in a good spot to be published.

Total _____ / 30

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True / False Questions [10]

True / False Question	Answer
Exploratory testing allows testers to learn, design, and execute tests at the same time.	**True**
A good feature specification should be vague to allow flexibility during testing.	**False**
Specification review meetings help identify missing or incorrect requirements before testing begins.	**True**
Unit testing is typically performed after system testing.	**True**
Black-box testing focuses on testing system behavior without examining internal code structure.	**True**
White-box testing examines internal program logic and code structure.	**True**
Error-case testing helps ensure the system handles unexpected inputs correctly.	**True**
UX testing only evaluates system performance and ignores usability.	**False**
Security testing includes checking authentication and authorization mechanisms.	**True**
Maintainability testing helps determine how easily a system can be monitored and updated.	**False**

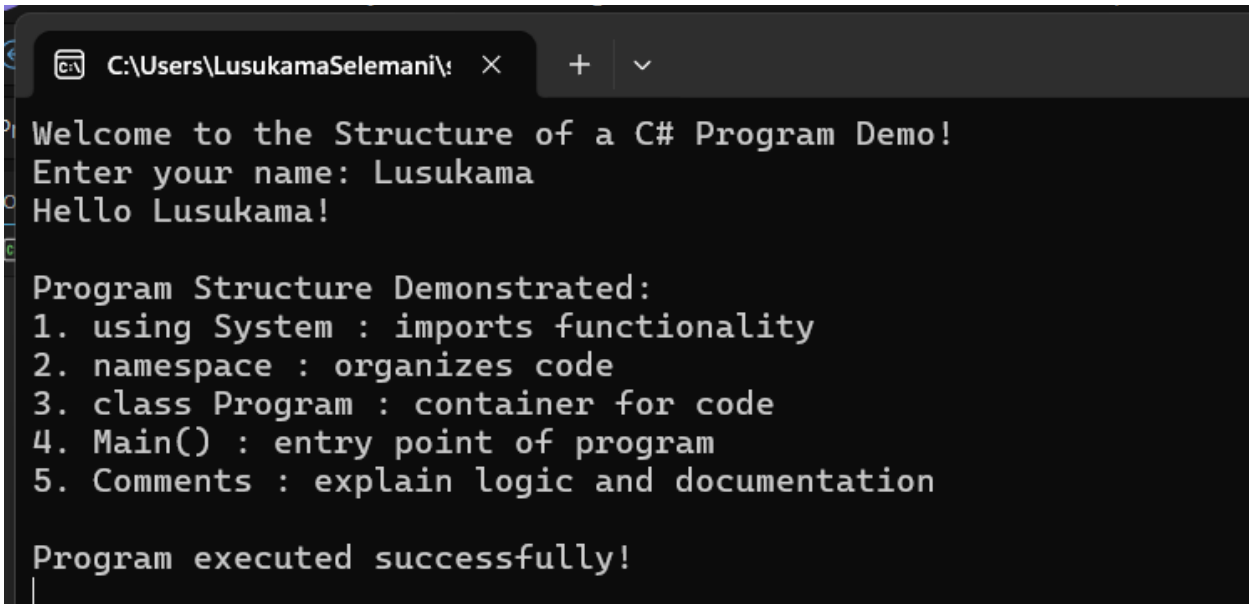
Total: _____/ 10

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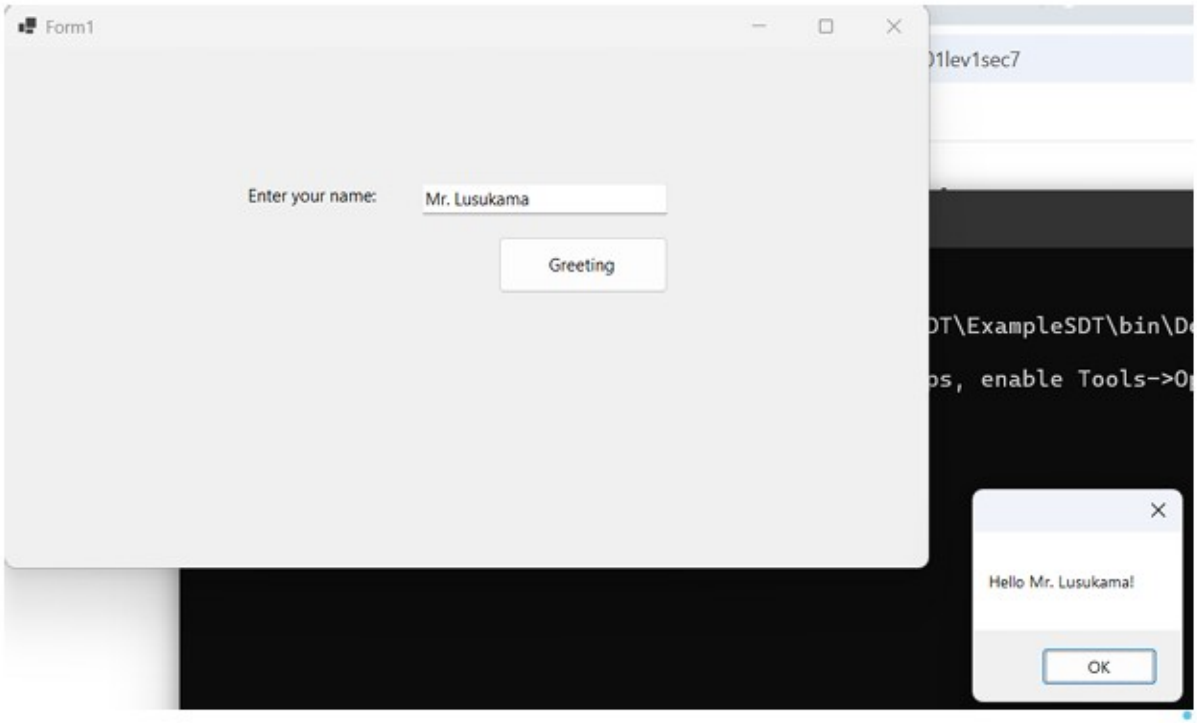
SECTION B: Short Practical Questions (10 Marks)

Question 1

1.1 Create a C# Console Application that the display the below output: (5)



1.2 Develop a Windows Forms Application that allows the user to enter their name and display a greeting message. (5)



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SECTION C: Scenario-Based Practical [50 Marks]

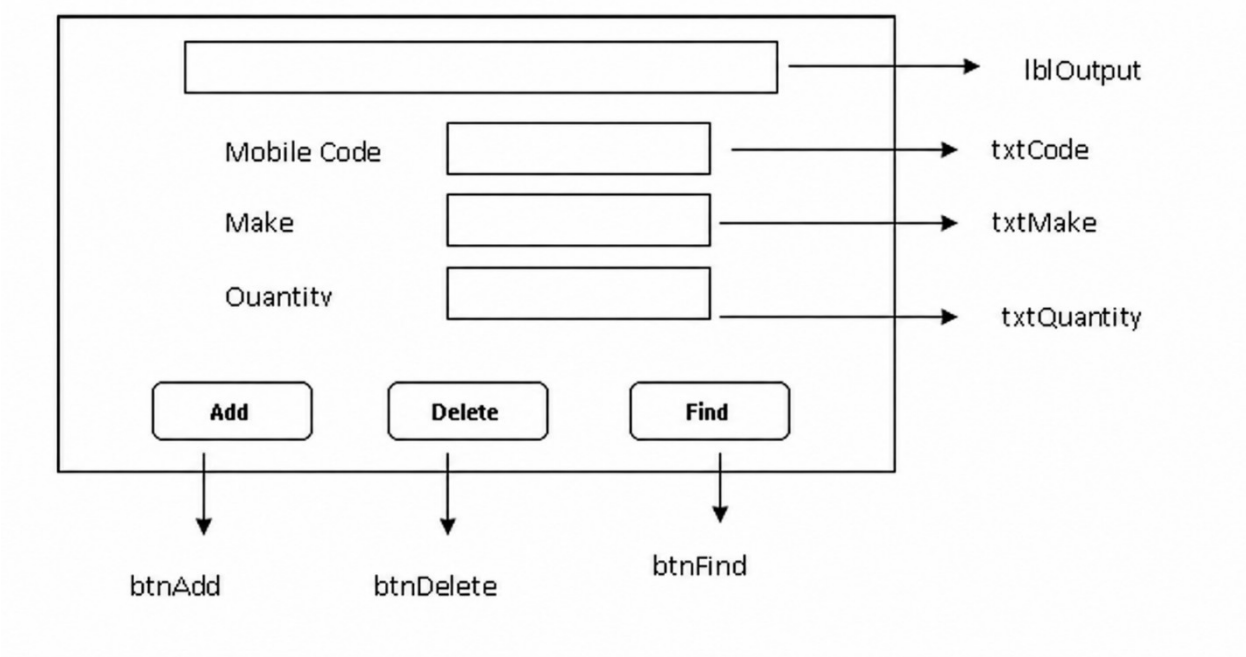
Question 1

You are working as a Junior Software Tester in a retail inventory company. The development team created a Mobile Stock Capture Windows Forms application, but the system contains interface naming issues, missing validations, and incorrect button behavior.

You are required to design, test, debug, and improve the Windows Forms application shown in the diagram.

SECTION C.1 — User Interface Design (10 Marks)

1.1 Create and design the following user interface using appropriate C# Windows Forms App controls and components. (10)



SECTION C.2 — Button Event Logic Implementation (20 Marks)

1.2 Write the events for each button

1.2.1 When the btnAdd button is clicked, the program should insert a record into the tblMobilePhones with the MobileCode, Make and Quantity from the corresponding data from the form and and display the message "Record Added" in the lblOutput. (8)

1.2.2 When the btnDelete button is clicked, the program should delete the record whose MobileCode is typed in the txtCode text field and display the message "Record Found" in the lblOutput; or "Record NOT Found" if the record is not found. (7)

1.2.3 When the btnFind is clicked, the program should find the record whose

MobileCode is typed in the txtCode text field and display the message "Record Deleted" in the lblOutput. (5)

Note: use functional and non-functional tests for this question

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SECTION C.3 : Software Testing Report (15 Marks)

1.3 Produce a Software Testing Report

Using the results from Question 1, compile a professional software testing report for the Mobile Stock Capture Application.

Your report must include the following sections:

1.3. 1 Application Description (3 Marks)

Provide a brief description of:

- Purpose of the application
- Main system functions
- User inputs captured

The purpose of this application is to keep track of mobile phone codes, the make of the phone and how much we have of the phone. This is basicly a way to keep track of all your inventory and a mobile phone store. The main system functions is to add, remove, and find mobile phones based on the code entered. User inputs captured are mobile phone codes, the make of the phone and how much of the phone.

1.3.2 Test Cases Table (5 Marks)

Create a table similar to the example below:

Test Case	Input	Expected Result	Actual Result	Status
Input Validation	Empty String	Popup prompting the user to enter something	Nothing came up	Fail
Add button	Add button	It should add the data to the table	It says record added on the output lable but the tables stays empty	Fail
Delete button	Delete button	It should delete the mobile code entered in the textbox from the table	The if statement didn't work	Fail
Find button	Find button	It should find me the record	It displays record deleted on the lbloutput	Fail
Input Validation	Making sure it saves the inputs	After adding a section of test code it to display the inputs	It does save the inputs as variables and displays them so the values are there.	Pass

Include at least five test cases.

1.3.3 Identified Defects (4 Marks)

List any problems found during testing:

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I've never worked with a table before so I don't know how to add or remove values from it. The only problems during testing was testing an incomplete program.

1.3.4. Recommendations for Improvement (3 Marks)

Save a local list or text file instead of the table for such a small program so that the data is permantly stored, add input validation to each textbox and make the program look nicer.

SECTION C.4 — GitHub Version Control Evidence (5 Marks)

1.3.5 Use of GitHub

To obtain the full 5 marks, learners must demonstrate correct use of GitHub version control by completing the following tasks:

- Create and work within a separate branch (not directly on the main branch)
- Make meaningful and professional commit messages that clearly describe the changes made
- Successfully merge the branch into the main (origin/main) branch after completing development
- Ensure the repository contains the complete working solution
- Submit a valid and accessible GitHub repository link

https://github.com/Unknown-S48/SDT621_WianSandenbergh_Summative_Assesment

End Exam

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SECTION A — Knowledge Questions (40 Marks)

Purpose of Section A

Assess theoretical understanding of testing techniques, validation strategies, and structured test documentation required before practical debugging begins.

QCTO Mapping

Question Area	Marks	QCTO Criterion	Evidence Demonstrated
Non-functional testing definition	2	AK0101	Knowledge of testing concepts
Load testing explanation	1	AK0101	Understanding performance testing
Load vs Stress testing comparison	2	AK0101	Distinguishing test strategies
Black-box vs White-box testing	3	AK0101	Test method classification
Equivalence partitioning	3	AK0101	Functional test design
Boundary value test cases	4	AK0101	Boundary testing application
Error-case testing definition	1	AK0101	Exception scenario awareness
Unexpected input validation importance	2	AK0101	Reliability assurance concepts
Login error-case examples	2	AK0101	Functional validation thinking
Definition of test case	2	AK0102	Test documentation knowledge
Components of test case	3	AK0102	Structured testing documentation
Test summary & stakeholder recommendation	5	AK0102	Test reporting interpretation

Section A Mapping Summary

Criterion	Marks
AK0101 Knowledge of testing techniques	25
AK0102 Knowledge of test report writing	15

Total Section A = 40 Marks

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SECTION B — Short Practical Questions (10 Marks)

Purpose of Section B

Assess ability to implement and validate basic working software functionality before applying structured testing.

QCTO Mapping Table

Task	Marks	QCTO Criterion	Evidence Demonstrated
Console output application	5	IAC0102	Correct functional output validation
Windows Forms greeting application	5	IAC0102	UI interaction testing

Section B Mapping Summary

Criterion	Marks
IAC0102 Correct tests applied	10

Total Section B = 10 Marks

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SECTION C — Scenario-Based Practical (50 Marks)

Purpose of Section C

Assess full PM-04 competency:

Test or debug source code to ensure client needs are met

Learners must design, debug, validate, test, and document results.

SECTION C.1 — User Interface Design (10 Marks)

Task	Marks	QCTO Criterion	Evidence Demonstrated
Correct UI controls selected	2	IAC0101	Functional test readiness
Correct control naming	2	IAC0101	Interface clarity
Layout alignment with specification	3	IAC0101	UI correctness verification
Proper component usage	3	IAC0101	Interface usability validation

Mapping Summary

Criterion	Marks
IAC0101 Functional & non-functional tests identified	10

SECTION C.2 — Button Event Logic Implementation (20 Marks)

Task	Marks	QCTO Criterion	Evidence Demonstrated
Add button functionality	8	IAC0102	Record insertion testing
Delete button functionality	7	IAC0102	Conditional logic validation
Find button functionality	5	IAC0102	Search behaviour verification

Mapping Summary

Criterion	Marks
IAC0102 Correct tests applied	20

SECTION C.3 — Software Testing Report (15 Marks)

Task	Marks	QCTO Criterion	Evidence Demonstrated
Application description	3	AK0102	Documentation understanding
Test cases table	5	IAC0102	Test execution structure

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Identified defects	4	IAC0101	Fault detection ability
Recommendations	3	IAC0103	Improvement reporting

SECTION C.4 — GitHub Version Control Evidence (5 Marks)

Task	Marks	QCTO Criterion	Evidence Demonstrated
Branch created and used	1	IAC0102	Version control workflow
Professional commit messages	1	IAC0102	Change tracking clarity
Multiple commits recorded	1	IAC0102	Development progression evidence
Branch merged into main	1	IAC0102	Integration workflow
Repository complete & accessible	1	IAC0103	Submission verification

SECTION C Mapping Summary

Criterion	Marks
IAC0101 Tests identified	14
IAC0102 Tests applied	26
IAC0103 Test report produced	10

Total Section C = 50 Marks

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FINAL OVERALL PM-04 ASSESSMENT SUMMARY

QCTO Criterion	Description	Marks
AK0101	Knowledge of testing techniques	25
AK0102	Knowledge of test reporting	18
IAC0101	Functional & non-functional tests identified	14
IAC0102	Correct tests applied	36
IAC0103	Test report produced	7

TOTAL = 100 MARKS

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